

$$\neg p \wedge (p \Rightarrow q)$$

$$\neg p \wedge (\neg p \vee q) \quad \leftarrow$$

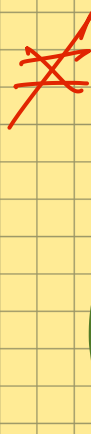
$$(\neg p \wedge \neg p) \vee (\neg p \wedge q)$$

$$\neg p \vee (\neg p \wedge q)$$

$$p \wedge (q \vee \neg q) \quad \uparrow \quad \text{True}$$

$$p \wedge \text{true}$$

$$p \quad \leftarrow$$



$$\neg(q \Rightarrow p) \quad \leftarrow$$

$$\neg(\neg q \vee p)$$

$$\neg(\neg q) \wedge \neg p$$

$$q \wedge \neg p \quad \leftarrow$$

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$$\neg p \Rightarrow (q \wedge \neg q) \quad \text{false}$$

$$\neg p \Rightarrow \text{false}$$

$$\neg(\neg p) \vee \text{false}$$

$$p \vee \text{false}$$

$$p \quad \leftarrow$$

$\forall \rightarrow$ for all

$\exists \rightarrow$ there exists